

What should the program do? (Program Requirements)

1. take in user input
2. compute the sum of all given input (*user_sum*)
3. calculate the total number/amount of numerical input provided (*total_input*)
4. calculate the average by dividing *user_sum* by *total_input*
5. output the correct average in all or a high majority of cases

What restraints are there? (Design Constraints)

The deadline for this task is 20 January 2026.

There is no language restriction.

I will write the program in Java.

A desired/required interface is not specified.

I will be using a CLI.

Functional Requirements

Input Format

What is the format of the input?

Valid data types include positive and negative integers, and positive and negative decimal values.

How should the data be stored?

User inputs will be stored in an array. This array is used to obtain two (2) instances of data:

1. The sum of the user's inputs (integers and decimals)
2. The total number of inputs provided by the user (positive integer)

The sum of the user's data will be assigned to a variable called *user_sum*. The total input provided by the user will be assigned to a variable called *total_input*.

Calculations

How will the program find the sum of all given numbers?

The user's input will be stored in an array. A variable, *user_sum*, will be assigned to hold the sum (initialized at 0) and the program will read through the array and add each number to the current value of the variable.

How will the program find the total amount/number of inputs provided?

Using the array that stores the user's input, the program will count each index of the array. This number will be assigned to a variable, *total_input*.

How will the program calculate the average?

The program will then be designed to divide the value of variable *user_sum* by the value of variable *total_input*.

Special Cases and Error Conditions

If a non-numerical value is provided, the program throws an error [specifying the problem] and the program resets.

If the user does not provide any input, i.e. the input is 'null,' the program throws an error that 0 cannot be in the denominator and the program resets.

If the user provides only whitespaces (spaces, tabs, new lines) as input, the whitespaces are replaced as null values. (the handling of null values is explained above).